

ANANTHU S AJAYAN

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SUMMARY

Machine Learning Engineer with hands-on experience fine-tuning LLMs, speech models, and vision models, building RAG systems, and developing end-to-end ML pipelines. Skilled in data preprocessing, model optimization, and deploying AI systems using PyTorch and Hugging Face. Brings 3+ years of combined experience in ML and backend engineering, including FastAPI/Flask, PostgreSQL, and cloud workflows on AWS/GCP. Seeking AI/ML roles focused on scalable model development and production-grade ML systems.

EDUCATION

MSc Artificial Intelligence

University of Southampton
Southampton, United Kingdom

Graduating Sept 2024
Merit

Relevant coursework: Machine Learning, Deep Learning Technologies, Natural Language Processing, Computer Vision

B.Tech in Computer Science and Engineering

TKM College of Engineering
Kerala, India

Graduating Sept 2020
8.00 CGPA

Relevant coursework: Data Structures & Algorithms, Machine Learning, Distributed Computing

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, Transformers, Scikit-learn, Deep Learning, NLP, RAG, MLflow

Programming & Backend: Python, FastAPI, Flask, REST APIs, PostgreSQL, MongoDB

DevOps & Cloud: Docker, Git, Kubernetes, Jenkins, AWS, GCP

PROFESSIONAL EXPERIENCE

Rootflo, Bangalore, Karnataka: AI/ML Engineer

Mar 2025 – Present

- Fine-tuned **LLMs**, **speech models**, and **CNN-based vision models** (ConvNeXt, EVA, DINOv3, etc.) for NLP, speech, and image tasks.
- Preprocessed and refactored complex datasets into standardized formats to improve model training stability and performance.
- Built **Document RAG** and **Image RAG** systems using embedding-based retrieval for knowledge-grounded responses.
- Evaluated RAG pipelines with **Ragas**, **DeepEval**, and custom metrics to assess retrieval quality and reduce hallucinations.
- Developed automated **ML pipelines** using **MLflow** for experiment tracking, metric logging, and model deployment.
- Designed and optimized machine learning models for various business use cases, including hyperparameter tuning and performance refinement.
- Worked extensively with **PyTorch** and **Hugging Face Transformers** for model training and fine-tuning.
- Built and maintained **FastAPI** services for backend features and ML inference endpoints, integrated with **PostgreSQL** for data workflows.
- Leveraged **AWS/GCP** cloud services and implemented asynchronous pipelines using **GCP Pub/Sub**.
- Collaborated with frontend teams to build interactive applications incorporating backend and ML-driven features.

Tata Elxsi, Trivandrum, Kerala: Python Developer

Jan 2021 – Feb 2023

- Built scalable **RESTful APIs** using **Flask** and integrated them with **MongoDB** for high-performance backend workflows.
- Worked with **Docker**, **Redis**, and **Celery** to support asynchronous processing and distributed task execution.
- Conducted API verification using **Postman** and maintained Git-based workflows using GitLab.
- Resolved backend issues during integration and contributed to deployment readiness.

- Automated internal workflows using Python and the **Robot Framework**.
- Developed web-scraping and data extraction utilities using **Selenium** and **BeautifulSoup**.
- Supported CI/CD automation and pipeline improvements using **Jenkins**.

ACADEMIC PROJECTS

Breast Cancer Detection using Federated Learning

June 2024 – Sept 2024

Implemented and evaluated federated learning strategies for privacy-preserving medical diagnosis.

- Designed and implemented **federated learning pipelines** to train a breast cancer detection model across distributed nodes.
- Compared federated approaches against traditional centralized machine learning models to analyze accuracy, robustness, and performance.

Real-Time Facial Emotion Recognition System

Aug 2019 – June 2020

Built a real-time facial emotion classification system using a Convolutional Neural Network (CNN).

- Developed and trained a **CNN** for real-time emotion recognition from video frames.
- Improved classification accuracy through advanced preprocessing, feature extraction, and model refinement techniques.

PUBLICATIONS

Real-Time Facial Emotion Recognition System with Improved Pre-processing and Feature Extraction

A. John, A. MC, A. S. Ajayan, S. Sanoop and V.R. Kumar, *"Real-Time Facial Emotion Recognition System with Improved Pre-processing and Feature Extraction,"* 2020 Third International Conference on Smart Systems and Inventive Technology (ICSSIT), 2020.